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CONSTIPATION AS EMBODIED OPPRESSION

Using Structural Competency in Community Health

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Introduction

Working in a high-income setting, constipation can seem simple and straightforward; physicians help patients connect their symptoms with the processed foods they eat and then discuss hydration, dietary modification, exercise, and efforts to wean medications such as opiates that can cause constipation. The view of constipation changes in resource-denied settings. This case study takes place at Jan Swasthya Sahyog (JSS), a community health center that trains and employs Village Health Workers (VHWs) to serve villagers in some of the most impoverished communities in rural Chhattisgarh, a state in Central India ([Jain 2016](#)). The lessons shared in this chapter were learned helping to train and guide VHWs caring for and triaging patients who complain of constipation and its associated complications—such as hemorrhoids, anal fissures and rectal prolapses—and from the women and men in the villages themselves. In these communities, it was clear that constipation was closely related to issues of extreme poverty, migratory labor, normalization of marginalization, and gender-based violence. In this chapter, we build on these insights to present constipation as a health problem that provides a window to various forms of social and structural causes of ill health. This is made visible through a structural competency analysis as a way to help us think through and act on the health impacts of constipation, including their deeper roots. This chapter explores the ways in which the entanglement of politics and healthcare can influence—and even cause—disease. We end with a multi-level approach to global health practice that responds to these causes and influences.

Structural competency

Structural competency is a concept developed by Jonathan Metzl and Helena Hansen, which they define as “the trained ability to discern how a host of issues defined clinically as symptoms, attitudes, or diseases... also represent the downstream implications of a number of upstream decisions about such matters as health care and food delivery systems, zoning laws, urban and rural infrastructures, medicalization, or even about the very definitions of illness and health” (Metzl and Hansen 2014). Their call builds on a long history of efforts to push back against hegemonic discourses in medicine that focus primarily on the individual patient, aiming to train physicians and other clinicians to think about health not just as an individual concern affecting the patient in front of them, but as the outcome of larger forces. Rudolf Virchow’s pioneering focus on social medicine in the 1800’s is one frequently cited example of this; other examples can be found around the world, including across Latin America, and in China, Japan, India, South Africa, and the United Kingdom (Pentecost et al. 2021). More recently, in the late 1990s and early 2000’s, the methods of cultural competence were developed to attempt this shift in the US on a cultural level. Observing the disjuncture of ideologies and epistemological (or knowledge) frameworks between physicians and patients from different cultural backgrounds, this intervention sought to provide cultural training to clinicians to help them understand the experiences and needs of their patients. Books like *The Spirit Catches You and You Fall Down* (Fadiman 1997), which tells the story of a Hmong girl with epilepsy living in Merced, California, and the well-intentioned but often conflicting efforts of her parents and doctors to care for her, provided a touchstone for this concept. Unfortunately, many of the resulting educational interventions quickly devolved into a list of stereotypical traits about different groups for physicians to commit to memory. Rather than bringing caregivers and patients closer together, these tactics tended to reproduce stereotypes and essentialism, or race- and ethnicity-based bias. Cultural humility attempted to attenuate this tendency, focusing on acknowledging the cultures of biomedicine and encouraging clinicians to consider their own social positions while committing to a life-long learning process (Tervalon and Murray-Garcia 1998). While problematic on a number of levels, cultural competence was quickly taken up on a curricular level, demonstrating the felt need for a more complex view of health and medicine within health professions training.

Structural competency proposes another attempt to rectify the failures of cultural competence. It aims to provide an important piece of health professions training, replacing cultural competence in curriculums, and bringing analytic tools from the social sciences to bear on the practice of medicine and public health. Most importantly, it shifts the framework away from blaming and stereotyping or essentializing culture toward recognizing how each person is embedded in a host of structural and institutional relations that condition their health, healthcare, and disease. Another key concept that structural competency builds on is the social determinants of

health. Studies by scholars such as Michael Marmot (Marmot et al. 2008) clearly demonstrate the direct relationship between socio-economic location and health, not just between the wealthiest and poorest, but at each level of the social gradient. Structural competency goes a step further, exploring the structural determinants of the social determinants of health. For instance, attending to the question of why poor people continue to be less healthy than their wealthier neighbors means also asking what social structures—including what policies and economic systems—are maintaining and reproducing these inequalities in both wealth and health. In this frame, the task of the caregiver is not to typologize patients but to be poised to be inquisitive about the social and structural positions and conditions of those they treat—as well as those that shape their own personal and professional lives.

Structural competency also draws on a number of key concepts covered in other chapters in this volume, including structural violence and vulnerability (see Chapter 1). Following the Structural Competency Working Group (SCWG) (Neff et al. 2019), we define social structures as “the policies, economic systems, and other institutions (policing and judicial systems, schools, etc.) that have produced and maintain social inequities and health disparities, often along the lines of social categories such as race, class, gender, and sexuality” (Neff et al. 2020). Structural vulnerability, a concept that builds on structural violence (Farmer et al. 2006), can be defined as “the risk that an individual experiences as a result of structural violence—including their location in socio-economic hierarchies” (Neff et al. 2020). Structural vulnerability can be understood through a range of mechanisms, including use of a structural vulnerability assessment (Bourgois et al. 2017). This tool has been used in a variety of clinical settings both within and outside the US and was used to structure the interviews conducted for this case study.

The case of cultural competence—and the long history of attempts around the world to make social and structural perspectives more central to health and healthcare, including current calls to decolonize medicine—provide an important and ongoing reminder to retain a critical lens, particularly in the operationalization of newer frameworks such as structural competency. Building on the work of cultural humility, this is referred to as “structural humility,” which Metzl and Hansen (2014) define as “the trained ability to recognize the limitation of structural competency.” When we work to integrate these perspectives into our work, it is important to maintain an active and critical view of how we understand both ‘the social’ (Adams et al. 2019) as well as the connections between our individual actions in a health setting and the larger structural conditions that shape both our lives and the lives of those we care for. For example, while potential reliance on a checklist signals the danger of replicating some of the damaging outcomes of cultural competence, here we emphasize its use as a tool for conducting in-depth interviews, rather than as a way to perform attention to a set of conditions without further engagement in their situated nature. Within the SCWG training, one implementation of structural competency, there is an emphasis on connecting this work through multiple levels of response (described after the case study) and situating

the work within organizing traditions such as the tradition of Beloved Community, core to the work of Martin Luther King, Jr. The training also provides a range of case studies that explore the intersection of healthcare and advocacy, from the health work of the Black Panther Party (Nelson 2011) to the success of the Integrated Soft Tissue Infection Service Clinic (Messac et al. 2013).

Work on structural competency continues to be expanded as new trainings are created for different fields, and research is done to explore how the concept can influence health and health professions training both within and beyond the clinic. It has also been adapted for global health education, drawing on experiences from training and work at JSS and around the world (Harvey et al. 2022). Building on these concepts and questions, we use a structural competency framework to think about and respond to the health challenges faced by women in Central India like Nikita, whose story we introduce below.

Health challenges in Central India

Chhattisgarh is a state in Central India with a population of 26 million, and among the worst health indicators in India (UNDP 2005). In 2009, 49 percent of people in rural Chhattisgarh lived below the state specific poverty line of 911 Indian Rupees (\$12.5 USD) per person per month (Planning Commission 2011), and 70 percent met the criteria for multidimensional poverty (UNDP 2005). Over 30 percent of the population in Chhattisgarh are from indigenous groups who identify as *Adivasis* and are referred to as scheduled tribes by the Indian State (Ministry of Tribal Affairs, Government of India 2014). Both in Chhattisgarh and across the country, *Adivasis* and other marginalized groups have worse health outcomes than the general population, particularly in rural areas (IIPS and ICF 2017).

Jan Swasthya Sahyog (translated as “People’s Health Support Group”) is a non-profit organization in rural Chhattisgarh founded in 2000 to provide quality, acceptable, and affordable healthcare to the people in rural Chhattisgarh and Madhya Pradesh (JSS 2015). JSS is organized in a three-tiered structure. First, JSS trains Village Health Workers (VHWs) from the villages they serve to support population health. Second, they run three subcenters where Senior Health Workers see patients and triage acute issues. Third, JSS runs a secondary care hospital that serves as a referral base. The hospital has half a dozen outpatient clinics, several inpatient wards with over 100 beds, an emergency room, an intensive care unit, labor ward, and operating theatres. This center serves as the training hub for VHWs, for a nursing school that recruits from local communities, and for a family medicine residency training program. Research conducted by JSS shows that the tribal groups they serve have higher incidence of undernutrition, tuberculosis, cancer, and severe hypertension compared to nontribal groups (Jain et al. 2015). The same JSS study showed that the subpopulation of “particularly vulnerable tribal groups” (PVTGs) experience even higher rates of tuberculosis and illnesses that require major surgery compared to other tribal groups and nontribal populations (Jain et al. 2015).

In order to understand why the patients described in this case study are poor, and the ways in which they are marginalized, it is important to understand structures of Indian society, particularly the caste system. The noted Indian scholar and politician B. R. Ambedkar offers one explanation for the origins of the caste system. He argues that Indian society was initially divided into four classes based on occupation: Brahmins (priests), Kshatriyas (warriors), Vaishyas (traders and merchants) and Shudras (artisans, farmers, and laborers). A person from a particular class could move to another social class (Ambedkar 1916). At some point in history, the Brahmin class transformed itself into a closed group to form a caste; subsequently, other groups enclosed themselves in similar ways. One key requirement that distinguishes a caste from a class group is endogamy, or marriage only within that group. This requirement is one reason behind a variety of social practices, including Sati Pratha (burning of a widow on her husband's funeral pyre), bans on widow remarriage and mistreatment of widows, and child marriages. The closed system gave rise to a social hierarchy, in which Brahmins were considered the highest caste and the Shudras were considered the lowest castes (Ambedkar 1916). While the origins of the fifth caste group, known as Dalits (meaning 'broken') or Untouchables (Hunashimarad 2014) is poorly understood, one theory posits the group may have been formed when people from two different castes married and were excommunicated from their villages and broader society. Dalits were not allowed to have any contact with the other caste groups, as they are considered outside (or lower in status) than those in a caste group. They have traditionally performed jobs such as manual scavenging and were described by the British during the colonial period as "depressed classes".

The tribal population in India are not a part of the Hindu caste system, nor are they counted as Dalits or untouchables. These indigenous groups, referred to as Adivasi (Xaxa 1999), are considered to be people dependent on forests for their livelihoods, less affected by the modernization of daily life seen elsewhere in India, and outside of the normative social hierarchy. Some outside scholars, such as Verrier Elwin (Guha 1996), claim that Adivasis are content with the ecology of the forest and not interested in outside systems of healthcare or education. When attempts have been made to integrate them into the dominant society, they have been displaced, separated from their lands and the ecosystem with which their livelihoods have been intertwined. While different groups have adapted to this in different ways, the social lives and structures that were developed in relationship to the lands that they traditionally lived on have been fundamentally altered.

As the British were handing over government administration after Indian independence, they were concerned about the status of marginalized populations in Indian society and created separate electorates for Muslims, Sikhs, Christians and other groups, as well as "depressed classes" and Dalits. Gandhi protested this structure, and it was replaced by a system of reserved constituencies for depressed classes based on schedules (formal lists of marginalized groups, with a certain number of government or other positions set aside for people in each group).

Because of this, what were considered “depressed classes” are today known as “scheduled castes” (SC). Similarly, Adivasis, or tribal groups, received reserved constituencies based on schedules, and are known as “scheduled tribes” (ST). A subset of tribes are considered “particularly vulnerable tribal groups” (PVTGs). SCs, STs, and PVTGs are administrative categories, and this affects a broad range of affirmative action programs, from admission to schools to government positions (Planning Commission 2003). The Indian constitution, while providing privileges in the form of reserved constituencies to the Adivasis, also disenfranchises the Adivasis of their ownership of the land, including access to the forests and animals that live there (Roy 2011), creating new forms of dependency and impoverishment.

The state of Chhattisgarh was formed from a portion of the large Central Indian state of Madhya Pradesh in 2000 and has the highest percentage of scheduled tribes living in any Indian state. While the land is rich in mineral resources, industrial development has not brought proportionate development for the people of Chhattisgarh in the 75 years since India’s independence from Britain (IFAD 2014). Land acquisition by the state for the setup of development projects such as mines and dams occurred in regions where mostly tribal people were settled, resulting in large-scale displacement. The majority of tribal populations depend on rural activity and are not familiar with the economic structures, politics, and cultural practices of mainstream Indian society. With the nationalization of forests and ongoing displacement, scheduled tribes have become more dependent on agriculture, and to a limited extent on unskilled jobs in urbanized areas (IFAD 2014).

A 2018 report of the Expert Committee on Tribal Health states that seven decades after Independence and despite many constitutional provisions to safeguard their interests, the tribal population continues to suffer disproportionately from health problems, compounded by problems of healthcare access and quality (Expert Committee on Tribal Health 2018). Adivasi groups across India, like most people around the world, utilize a range of systems to understand and respond to health and illness, a practice referred to as “medical pluralism.” Anthropological studies report that members of the *Gond* tribe, which this case study focuses on, utilize a combination of treatment based on herbs and plants native to their region, diagnosis and treatment by religious healers, and interaction with the biomedical system through both public and private providers and clinics (Tripathi, Preetha and Vijay 2024; Xaxa 2014). However, the public health system remains inadequate in the tribal areas, including limited infrastructure, human resources, and governance. In addition, the current public biomedical healthcare delivery system does not account for the other healing systems many Adivasi people follow, leading to low rates of acceptance. Because of this, the organization of public health service delivery in tribal areas suffers from serious design flaws that affect its efficiency, efficacy and uptake.

In response to these gaps in care, the clinical space and care provided at JSS have been designed specifically to serve the people living in the surrounding area.

JSS is situated near the Achanakmar Tiger Reserve, which is home to Adivasi people, primarily the Gond and *Baiga* Adivasi groups. The Reserve was created in 2009, in order to preserve the forest and the tiger population. However, the creation of the reserve and the Indian Forest Act ([The Indian Forest Act 1927](#)) meant that many of the tribal groups who lived in the forest have been forced to leave and adapt to life in cities ([Shrivastava 2017](#)) and towns ([Fanari and Paliwal 2017](#)). When JSS was built, locations for sub-centers were chosen to be close to potential patients who have been able to remain in their home districts. Specifically, centers were situated inside the forest to be more inclusive of the needs of the Adivasi population of Achanakmar Tiger Reserve. In addition, the physical structures at JSS were built to be as familiar as possible to Adivasi communities. None of the buildings on campus are more than one story above the ground floor. The bricks and walls are not painted, and the roof is constructed in the same way as the roofs in villages. Trees have been planted in the hospital grounds, and waiting rooms are provided for relatives to stay while the patient is admitted. The food provided on the campus is similar to the food patients and their families might make and eat at home, and herbal medicines are included in the formulary to treat appropriate ailments. In addition, JSS seeks to minimize financial barriers by offering high quality but heavily subsidized—and often free—care to patients who are otherwise unable to afford care. Finally, the health workforce at the hospital consists predominantly of people from the surrounding community. To support this, a nursing school run by JSS on the same campus admits only Adivasi women from the district in which it is located, and from one adjacent district. This is part of an ongoing effort to provide care to women like Nikita, who we meet in the next section.

Case study: Nikita's constipation

Constipation is an extremely common medical issue. It is difficult to define precisely but includes low number of bowel movements per week, excessively difficult defecation, and hard consistency of stools. Chronic constipation results from inadequate fiber or fluid intake, or from disordered colonic transit or anorectal function which in turn have a host of underlying causes (including, in some cases, from medications). If dangerous warning symptoms or signs in older patients (such as weight loss or anemia) are noted, colonoscopy is often recommended to evaluate for signs of cancer. Rectal examination is needed to evaluate the patient for hemorrhoids, anal fissures, fecal impaction and features of evacuation disorders. In most cases, an underlying cause is not found, and patients respond with hydration, exercise, and dietary fiber supplementation. Further treatment involves pelvic floor training exercises, bowel regimens of laxatives, stool softeners, suppositories, or enemas as needed ([Kasper et al. 2015](#)).

Both hemorrhoids and anal fissures are complications from constipation. Symptomatic hemorrhoids, swollen veins in the rectum and anus, occur due to straining

and constipation. This can cause bleeding or pain. Anal fissures can be caused by trauma to the anal canal following defecation, irritation from the trauma resulting in increased sphincter tone, relative ischemia and poor healing. This causes severe pain with defecation. Treatment involves sitz baths (warm shallow water), increased dietary fiber and stool softeners. In the case of hemorrhoids there are several clinical procedures which can be done if these fail. In the case of anal fissures, topical anesthetics and ointments that lower the anal tone are often also prescribed. If these measures fail, surgical procedures can be offered ([Kasper et al. 2015](#)).

In 1993, Ramanakumar et. al. found that indigestion, gas, acidity, and constipation were the second leading causes of morbidity among rural women in India ([Ramanakumar 2004](#)). These statistics are born out in the lives of patients who come to JSS. In one clinic session, out of an average of 40 patients, over half needed evaluation for constipation-related issues. Many had anal fissures and external hemorrhoids. JSS and a visiting global health equity fellow with the HEAL initiative worked together to collect data from each sub-center, using door-to-door surveys confirmed during clinical evaluation (see the introductory and conclusion chapters of this volume for more information, and [HEAL 2018](#)). Senior Village Health Workers brought in all patients who responded to having symptoms of constipation. Based on data collected over the course of three weeks, it was found that approximately 40, 23, and 45 percent of female patients from Bahmni, Shivtarai, and Semarya subcenters respectively suffered from constipation. This sample included children ages two and older. Of the patients reporting constipation, approximately 26, 17, and 25 percent respectively suffered from either external or internal hemorrhoids and 14, six, and 22 percent respectively were found to have anal fissures. These numbers are likely under-reported, as many patients refused to take part in the initial survey, fearful they would have an examination.

Nikita's story

The story of a woman we will call Nikita helps to illustrate both the health challenges and structural causes of constipation that these numbers reflect in rural Central India. Nikita was 20 years old, from the Gond tribe, and came to the JSS sub-center in the town of Shivtarai, concerned about rectal bleeding. About two and a half years prior, she had started to notice that she had to strain to go to the bathroom. Then she noticed that it became more and more painful to pass stool. As time passed, she noticed rectal bleeding when she went to the bathroom. Increasingly, each bowel movement took more effort and became excruciatingly painful. This frightened her, but she felt too ashamed to tell anyone what she was experiencing. Six months later, she finally talked to her grandmother. Her grandmother sent her to the traditional healer, who prescribed herbal medicines. The medicine tasted awful but seemed to help; she no longer saw any blood when she moved her bowels, and they were less painful. She stopped taking the medicines, and within a few months the problem returned. She mustered the courage to talk to a Village Health Worker

about it. The health worker brought her to a health camp focused on constipation. The doctor at the camp listened to her history, examined her and found an anal fissure and external hemorrhoids, both signs of longstanding severe constipation. The doctor told Nikita she had hemorrhoids and gave her medicines. Nikita returned for follow-up at one of the subcenters where a Senior Health Worker (SHW) and social scientist interviewed her. The SHW started discussing prevention tactics and providing guidance to help with constipation, but Nikita complained it was not fiber, food, or hydration that was the problem. She told the SHW that she deliberately refused to evacuate her bowels for days on end. This prompted the SHW to ask Nikita to share more about her life.

Nikita was enrolled in school through the tenth grade but did not complete her secondary education. She was married, but her husband had migrated to Gujarat for work, hundreds of kilometers away. She lived with her in-laws and their three children and grandchildren, a total of ten people living together in a mud house with a scaffolding roof. Her family owned two acres of land which they farmed, but this land did not yield nearly enough to sustain them; at least five acres is needed to sustain a small family in rural India (Yogesh Jain, informal discussions with authors, 2017). Consequently, most of the family were enrolled in the government National Rural Employment Guarantee Act (NREGA) scheme, which guarantees every adult in rural India 100 days of unskilled paid work ([The Mahatma Gandhi National Rural Employment Guarantee Act 2005](#)). One year, Nikita's pay was delayed, although everyone else in her family received their money. Her father-in-law and brother-in-law handled the family finances, so she was not sure why her pay was delayed. The same year, her father-in-law was diagnosed with tuberculosis and had to take medicines for nine months. Money was scarce, and they took out a loan. Since her brother-in-law had started working as a mason, most of that loan had been paid off, and the family was doing a lot better. In addition to working their own farm, and jobs through the NREGA scheme, they worked in the fields of other farms. They had not been able to sell any of their produce, because they needed it all for subsistence. The SHW asked Nikita about her diet, a key determinant of constipation. Her family got some food through ration cards and the public distribution system (PDS), but it was not enough. For her family of ten, they only got 28 kilograms of rice; they should have gotten at least twice this, but her brother-in-law's family had not yet received their social security card (*Aadhar*), and so they had not been able to link their ration cards with their Aadhar IDs. In addition to the rice, PDS also included a small amount of chickpeas, salt, and sugar. In the winter months, her sister-in-law and mother-in-law would go to the forest to forage for vegetables, but otherwise potatoes were the only vegetable they could afford.

The SHW asked her about how she stayed hydrated, another key aspect of preventing constipation. There was a community hand pump near her house, so hydration was not an issue for Nikita. She was responsible for getting water from the community hand pump both for drinking and other uses, including for flushing the latrine. The family used about 200 liters of water per day, and she carried 25 liters

(which weighs over 50 pounds) on each trip back from the pump. Before her family had a latrine, she was careful not to drink too much water, so she would not have to go to the bathroom often. She does not consciously make the same decision now. Before the family had access to a latrine, she also practiced open defecation; she did the same when she was working out in the fields. Nikita explained how in places like Shivtarai, women do not have the opportunity to use a latrine. There was often no latrine, or it was unsafe. Village boys would often hang around the latrine and make lewd comments, which would make her uncomfortable. So Nikita often held her stools until she had enough time to walk to the fields to relieve herself. To reach the fields she had to walk long distances, one to two kilometers away from her home. If there was more household work or farming to be done, she would stall and wait until the next day, or sometimes a few days after that. When she did go, she would usually travel with one of her female family members, and together they walked into the forest, but not deep enough to be at risk of running into wild animals. Her sister was afraid of scorpions and snakes, but Nikita had been lucky and had not been bitten. It was important to go with somebody else because gangs of boys, usually drunk, often harassed women in the streets. The villagers were even more afraid of the police than the drunk boys so there were no repercussions for the boys. Because of this, Nikita never traveled alone.

Two years ago, the government launched a new scheme, *Swachh Bharat Abhiyan*, to build latrines in every home. The villagers were not consulted, but most did not object. Nikita's family now had a latrine in a pink outhouse in their yard and she had been able to use the toilet about twice a day. Surprisingly, even though she now used the latrine, she had not seen much improvement in her hemorrhoids. Instead, she came to associate using the toilet with pain, since all her defecations had been painful. At the constipation health camp, the doctor asked some questions and examined her but seemed to be in a hurry. The doctor gave Nikita some medicines and told her that with treatment, her hemorrhoids and constipation would improve. The doctor did not explain much else and went off to see the next patient. Nikita was glad she did not have to go to the main hospital; she had always been afraid of leaving her village and had never done so.

Constipation in Chhattisgarh: A structural health condition

Nikita is not alone. Extreme poverty, migratory labor, and gender-based violence are among the many challenges faced by the rural poor in India. Due to economic conditions, the Adivasi women of central rural Chhattisgarh are taking on increasing farming responsibilities beyond the daily routine tasks traditionally imposed on women: cooking, cleaning, and taking care of the home. Like Nikita, many women are suffering from constipation and its aftermath. How do the conditions of social inequality, government-authorized disenfranchisement, gender inequality, and poverty help us to understand the high rates of constipation in Chhattisgarh?

In interviews with patients about their constipation and its sequelae, several themes emerged: the risk of gender-based violence, the conditions of poverty and resource deprivation, and the politics of open defecation. We learned it was common for women to be afraid of traveling alone to remote areas away from the village because of the risk of violence. Like Nikita, many women have heard stories or known young women within their own and other communities who have been sexually assaulted. This phenomenon is widespread in rural India. For example, a study of women across the life-course in the neighboring state of Odisha found that environmental barriers, social factors, and fears of sexual violence all contributed to sanitation-related psychosocial stress (Sahoo et al. 2015). Though women responded with small changes to sanitation practices, they were unable to significantly modify their circumstances, notably by achieving adequate safety and privacy for sanitation-related behaviors. This also means that despite the efforts to build latrines in villages, women may be the least likely to benefit from them because of larger socio-political gender-based inequality.

Moreover, the 2011 Indian census data shows that seventy percent of rural Indians did not have a latrine (Government of India 2011), and India has higher rates of open defecation than countries with lower incomes and educational attainment, including Bangladesh, Democratic Republic of Congo, and Malawi (Coffey et al. 2014, 2017). Many global health practitioners coming from the US and Europe have relied on the belief that latrines are the safest and most efficient solution to open defecation. Why do women continue to openly defecate despite increased constipation, anal fissures, and hemorrhoids caused by the need to wait and travel in groups? A survey in rural north India about Sanitation Quality, Use, Access, and Trends (SQUAT survey) provided several explanations (Coffey et al. 2014, 2017). First, some women reported greater fear of using a latrine. In the SQUAT survey, just over four percent of women reported they were victims of an attempt to molest them while going to defecate (Coffey et al. 2017). Open defecation allows for vigilance and surveillance.

Second, problems were reported related to pit latrine design and notions of purity in society. WHO-recommended pit latrines have played a major role in eliminating open defecation in developing countries and have been endorsed for use by the Indian government in the Swachh Bharat Mission (Coffey et al. 2017). However, people wrongly assume the latrines will fill up quickly and will require emptying, which is particularly problematic as manual cleaning of human feces is considered the most degrading of tasks and has traditionally been assigned to the lowest Dalit castes (Coffey et al. 2017). Thus, rural Indians associate pit latrines with pit emptying and untouchability and find these affordable latrines to be physically dirty and ritually polluting (Coffey et al. 2017). Evidence from the SQUAT study shows that 48 percent of “households with a working latrine... had at least one member who nevertheless defecates in the open” (Coffey et al. 2014). This proportion was higher amongst households with a government built latrine than privately built latrine (Coffey et al. 2014), which are larger and less likely to need to be emptied in a lifetime.

Third, participants reported a desire and need for open defecation, stating that when they did go to stool outdoors, they felt greater satisfaction. This is again supported by evidence from the SQUAT study which showed that amongst those who practice open defecation, 47 percent explain that they see it as “pleasurable, comfortable or convenient” (Coffey et al. 2014). It is also seen “as promoting purity and strength” (Coffey et al. 2017). Finally, women reported that it allowed them to leave home and provided an “opportunity to take a morning walk, see their fields, and take in the fresh air” (Coffey et al. 2014, 2017). Overall, latrine policies failed to recognize the health effects of open defecation, (Coffey et al. 2014) as well as the health and social risks of latrine use. Open defecation may be preferable, but it can also be dangerous, promoting long wait times for defecating and other risks. These findings point to the contradictions in defecation politics, and to the importance of recognizing the larger forms of structural violence that impact the health and social risks of both enclosed and open options.

Through Nikita’s story and the experiences shared by others, we learned about the ways poverty and women’s workloads mean they often do not have time to go to the bathroom, that gender-related safety issues around latrines are a major concern, and that to stool safely they often travel far distances. These problems are tangled with other larger concerns related to the government’s efforts to “modernize” Adivasi people, compelling them to use latrines as a matter of self-care, but failing to recognize the social constraints against using them. Using a structural analysis framework allowed us to make explicit the various social structures that were causing suffering, the ways that such structural violence was being perpetrated, and potential pathways for intervention (see Figure 4.1). From this analysis the structural barriers became clear: marginalization of indigenous communities, limited funding of social services and safety nets, and gender-based violence. Using the analytic tools described in this chapter, the authors then started a conversation about solutions to address these underlying structural problems.

Insights for critical global health practice

After hearing many stories like Nikita’s, doctors, program coordinators, and Senior and Village Health Workers at JSS discussed the issue of constipation and its health impacts. A cultural competence framework deployed here might have focused on Nikita’s use of herbal remedies as a cause of the delay in her seeking biomedical care, or on a “culture” of open defecation. While JSS utilizes a form of cultural humility in integrating herbal formularies and familiar architecture on their campus, learning from the communities they serve, staying focused on this level of analysis would address only a small portion of the burden that Nikita is facing. To address these larger burdens, the JSS team started by thinking about the social determinants of health, with which they were familiar; poverty underlies most of the health conditions that JSS treats. Taking a step further back, they outlined the structural determinants of constipation

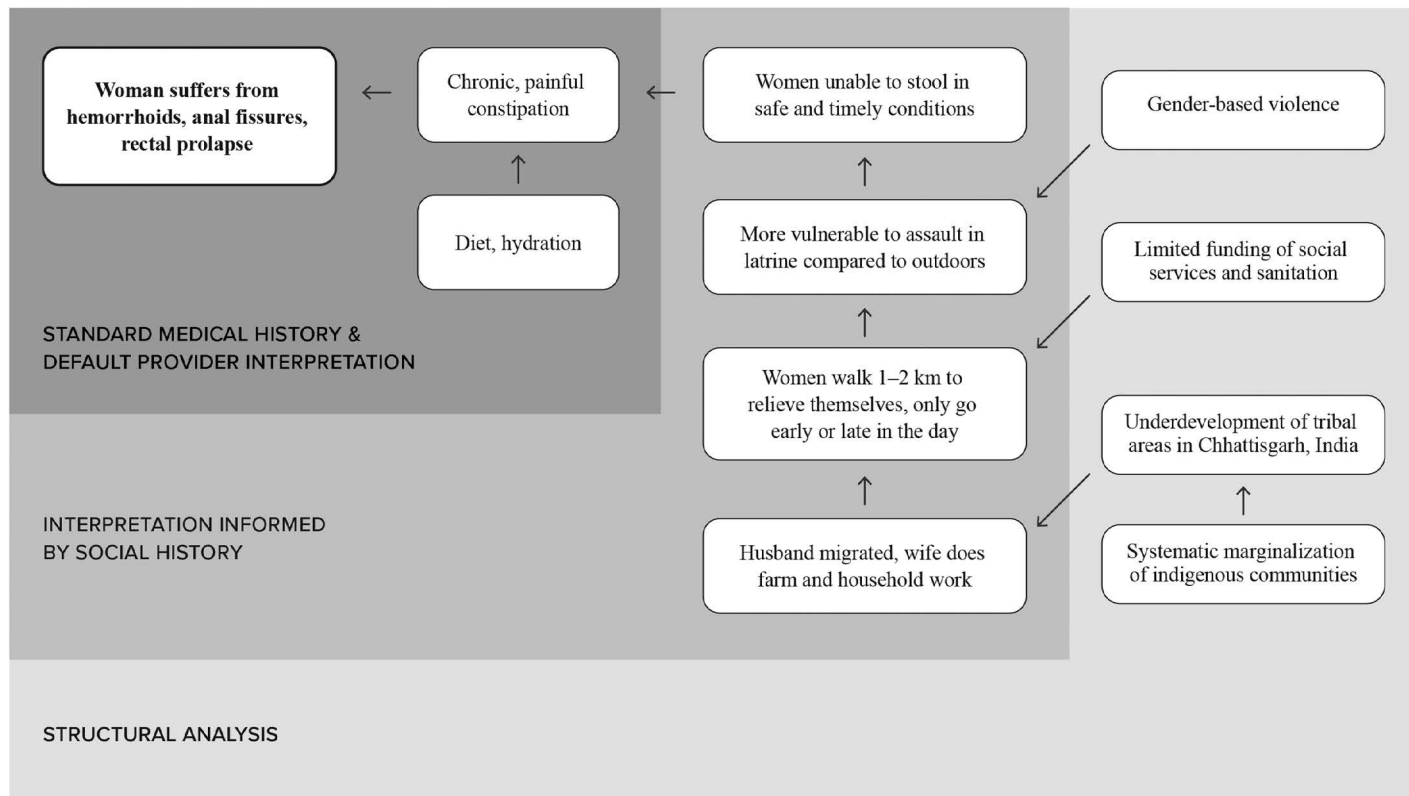


FIGURE 4.1 Flow chart illustrating a case like Nikita's, adapted in collaboration with the Structural Competency Working Group, and used in trainings (Neff et al. 2020).

in their patient population. They started by noting gender-based violence. This is a widespread and widely discussed problem; harassment of women walking alone, called “Eve-teasing,” is only the publicly visible side of a larger trend of violence against women. This is compounded by the lack of availability of toilets in schools and facilities to change and discard menstrual pads, one reason so many young girls leave school early, further exacerbating the gender gap in education, opportunities, and empowerment (Yogesh Jain, email correspondence with authors, 2017). Next, they discussed limited funding of social services and sanitation. The national government provided a number of latrines through their sanitation program, but this program was heavily top-down in design and implementation and use of the latrines was uneven. Chhattisgarh, which has a significant tribal population, was created as a separate state in November 2000, in part because of chronic underinvestment at a state level in services to that area. While there has been some improvement in services, and innovative programs such as the Mitadin community health worker program (based in part on JSS’s work, and in turn part of the inspiration for the national government’s ASHA community health worker program), the state has limited reach and limited funding. The ongoing underdevelopment of tribal areas, itself a structural concern, has also been shaped by the systematic marginalization of indigenous communities in India and around the world. Having arrived at this analysis, what can be done?

Too often in clinical medicine—and in global health more broadly—we fail to acknowledge and address the socio-cultural and political contexts of patients’ lives, often assuming they are beyond the reach of our clinical or programmatic capacities. Working together as clinicians and social scientists of global health to use a structural competency approach to this case changed the authors’ attitudes and provided us with tools to help improve clinical practice and to see the value of social science research and multi-level advocacy. Here, we share two tools that have been developed to help with identifying and responding to structural forces as they impact health—including even biological phenomena like constipation as a byproduct of larger socio-political forces. First, a structural vulnerability assessment was used to record social history. Second, we offer a tool to identify different levels of interventions and to brainstorm ways to move forward. Each tool is intended to be utilized critically, taking care not to use them to reify individualistic response to health, but to work towards a more structural response.

Tool 1: Structural vulnerability assessment to help record social history

In clinical medicine, supplemental laboratory testing and imaging technologies are used to help establish a diagnosis and tailor treatment. For example, blood tests can help establish the cause of anemia, and if it is identified as iron deficiency, then

increased iron can be prescribed. A CT scan can be used to identify a blood clot in the lung, after which patients are treated with blood thinners. However, the standard medical history approach gives a very narrow view of the problems that women like Nikita face. The therapeutic interventions that follow may help but often do not address underlying problems that cause constipation for this segment of the population. It is only after hearing more about Nikita's life—learning about her education, home, family, living quarters, financial constraints, food and water insecurity, the limited availability of latrines, and concerns about safety and gender-based violence—that we understood why many women in rural India are constipated.

While taking such extensive social history may seem daunting, a structural vulnerability assessment tool has been developed, tested, and used in a variety of settings to facilitate this process (Bourgois et al. 2017). We adapted the tool (see [Appendix A](#)) for use at JSS to guide semi-structured patient interviews conducted by providers. In US clinical settings, the authors have also seen an increase in the use of social history (sometimes referred to as social needs or social risk) in clinical evaluations (Martinez and Alper 2019). Nurses and physicians may select from options in an Electronic Health Record, for example asking a question about whether there has been a time in the past year when the patient could not pay their rent or mortgage, with options to answer yes, no, or decline to answer, or whether the patient has been worried that they would run out of food and not have enough money to purchase more, with options to answer “never true,” “sometimes true,” “often true,” “patient unable to answer,” or “patient declined.” While this is an important starting point, and some clinicians do use it to begin a longer conversation that can help to identify additional needs for support, this structure provides limited opportunities for discussion, or to enter more complex answers into the health record. In this case study, the structural vulnerability assessment was adapted to the context, based on the experiences of the clinicians, their colleagues, and patients—for example, when asking about access to affordable food, following up with specific questions about ration cards from the government both by name, and by the color of different types of cards. The goal was to gain a holistic view of patients' lives and barriers to health, both for this project, and as part of the larger health center's ongoing efforts to provide appropriate care and support to the community.

The tool was helpful in bringing out a number of themes that may not have otherwise emerged. Beyond enumerating social forces, the tool helped practitioners to reflect on patient attitudes and behaviors during the interviews. These attitudes and behaviors—and the way practitioners respond to them—should be considered among the social forces that impact health inequities. For example, it was noted that there was what is perceived as a “culture of silence” among the Adivasi population and that it was very difficult to engage in a long conversation with interviewees, a dynamic attributed to the historically paternalistic structure of the doctor-patient relationship (Bhavna Seth, email communication to Seth Holmes on “Structural Vulnerability Insights: Rural Central India,” 2018). As a result, the same question would have to be asked different times in different ways to elicit a

response and the interviews were very long, in some cases lasting up to two hours (Bhavna Seth, email communication to Seth Holmes on “Structural Vulnerability Insights: Rural Central India,” 2018). In this setting, the tool no longer functioned as a quick screening tool since it took much longer than anticipated to administer. Rather than functioning as a checklist that would provide quick and easy answers to large and complex problems, it provided a way for clinicians to better understand both the lives of their patients, and the larger social structures that lead to illness (Bosk et al. 2009).

Another important theme that emerged from the use of the tool was the lack of entitlement felt by patients. The majority of the patients interviewed had been deprived of basic human necessities—they were malnourished, lived in conditions of extreme poverty and had to travel long distances to access basic medical services. However, in interviews patients did not affirm the idea that they were discriminated against (Bhavna Seth, email communication to Seth Holmes on ‘Structural Vulnerability Insights: rural Central India,’ 2018). They expressed gratitude for the public distribution system, the free education system, and the health system, viewing these basic necessities not as rights but rather as luxuries. Their low expectations of services and resources from the national, state, and local government could be one factor related to why women avoided reporting harassment and other gender-based violence to police officials.

We also found that patients seemed to have a limited understanding of the concept of discrimination. One participant reported that she did not have access to sufficient food to eat, safe drinking water, education, or quality healthcare. However, when asked whether she felt discriminated against, she said “no.” Patients were aware of basic structures of social inequality, especially those based on caste, reflecting analysis far beyond a fatalistic acceptance of social inequality. As noted above, due to the paternalistic patient-doctor relationship, patients may have been reluctant to talk about the challenges of their living conditions in front of health workers to avoid appearing ungrateful. In addition, the conditions of inequality were so pervasive and ubiquitous that even if they were recognized as unfair, they were not seen as valid topics for complaint because they were perceived as “normal”—the conditions of suffering were themselves in some sense products of normalized or naturalized inequality (Bhavna Seth, email communication to Seth Holmes on “Structural Vulnerability Insights: rural Central India,” 2018, Holmes 2007).

Overall, we found that using the adapted version of the structural vulnerability assessment tool helped us to see beyond the biological body, and to recognize modes of intervention that were aimed at improving the conditions that produce Nikita’s constipation and sequelae, even though this meant attending to social and structural conditions that felt, at first, beyond the scope of clinical care. We encourage readers to use and adapt the tool for your own work, to pay attention to how you are using it, to how patients respond, and to how it might be changed or used differently to help support practitioners and patients to work together to address structural causes of health inequities.

Tool 2: Identifying levels of interventions

This case study was developed based on the premise that taking time to listen, explain, visualize, and invest in human connection, using the power women already have, will lead to more structurally competent outcomes compared to pursuing initiatives that do not reflect the needs and experiences of rural women. In the long run, we hope this can be part of a solution to help to sort out the challenges these women encounter in their lives. Whitehead and colleagues propose one typology for actions to address social inequities: strengthening individuals, strengthening communities, improving living and working conditions, and promoting healthy macro-policies (Whitehead 2007). Similarly, according to the Social Ecological Framework, interventions on identified social forces can be classified according to one of six different levels: intrapersonal factors, interpersonal processes and primary groups, institutional factors, community factors, and public policy (McLeroy et al. 1988). The Structural Competency Working Group built on these types of efforts to develop a tool for considering levels of intervention, considering intrapersonal, interpersonal, clinical, community, research, and policy interventions (Neff et al. 2019). Here, we explore these levels of intervention, and the ways that JSS has and can take steps at each to address health inequities, including those connected to constipation and related health outcomes. We also note the limits of this case study and analysis at a number of these levels.

Level I: Intrapersonal

At an intrapersonal level, the focus on Nikita's case led to time set aside for health workers to consider the health of their own bowels. During Village Health Worker (VHW) training sessions at JSS, the team decided to take a fresh approach to supporting women to understand their own bodies. Staff and HEAL fellows took the time to educate patients about the why's and how's of their own bowel ailments. Many of the women suffering from complications of constipation were VHWs themselves. Staff asked VHWs to assess one another and ask specific questions: do you have abdominal pain? What is your stooling frequency? What do you consider normal as regards stooling? Do you have hard stools? VHWs reported that by engaging in conversation about a common medical problem, they felt there was increased opportunity to connect and understand their own bodies and social situations. Questioning and diagnosing in turn led to physical examinations of one another. Many of the women were able to see the problems that chronic constipation can lead to, in themselves and in their colleagues—leading to interventions at the second level: interpersonal.

Level II: Interpersonal

After this training, VHWs were more equipped to perform rectal examinations and ask patients about pain, blood, and other bowel health symptoms, with VHWs conducting physical examinations for their own patients at the village level. The role of VHWs as members of the villages, with close connections to the social fabric, is by

design. JSS's strategy in building rapport and working to understand these women involved taking time to explain latrines, then helping them apply their own questions, thoughts, and feelings about stooling to their own colleagues and patients. Providers at JSS are trained to make sure that they are ready to approach each patient in a humanizing, caring way. By empowering women from the villages, who face the same challenges as their neighbors, JSS hopes to effect change in gender-based power dynamics at the household level. This is especially important because much clinical advice occurs at this stage—outside the clinic. Many patients were initially advised to increase their fiber intake and hydration to and try ayurvedic treatments, such as *avla* and *seral churna*. If these did not work or more complications were involved, such as rectal bleeding, patients were advised to come to JSS sub-centers for further evaluation by health professionals. Assessments at the sub-centers brought in over 30 patients to the main JSS hospital with various constipation-related complications, from significant external hemorrhoids to rectal prolapses. Bringing patients in from home visits and sub-centers to the JSS main campus leads to the third level of intervention: the clinic.

Level III: Clinic

The work of the VHWs provides care both outside of the clinic and brings patients into the main JSS campus for treatment when needed. Nikita, like many of the patients that JSS sees, interacted with a variety of care workers who provided a range of healthcare. After other remedies had not worked, a VHW brought her to a doctor at a health camp, who provided medicines. She then went for follow-up care at a subcenter, where a Senior Health Worker and social scientist interviewed her—illuminating the limitations of what this clinical care could do for her. While both herbal remedies and medications and procedures from the doctor and Senior Health Workers helped relieve her constipation and related symptoms in the short term and provided information about how to reduce and prevent constipation under ideal circumstances, this level of care was not able to intervene on the root causes of her health challenges. As is often the case, interventions past this level take time and have not yet led to changes that benefit Nikita directly.

Level IV: Community

Through the assessments and examinations described in this case study, a doctor at JSS asked VHWs what should be done to make conditions safe for them to stool. Many stated they just wanted proximity to home and safety. They did not want to have to walk long distances and did not want to worry about anyone assaulting them. This is an ongoing challenge, and while it is part of larger social change efforts that JSS is part of, it is not directly addressed by the actions described in this case study. Examples of potential interventions at this level include working with community groups and other NGOs to change gender-based power dynamics at the

community level, to build and maintain latrines in communities that are designed in locations and styles that meet the needs of community members, and working with law enforcement to ensure safe communities while recognizing the ways in which many people do not trust law enforcement.

Level V: Research

Research allows us to gather information about a range of experiences—in a community, a state, a country, or around the world—and analyze those experiences to help develop new understandings and interventions. JSS runs several research programs that seek to understand the experience of patients living in poverty in rural Central India ([Jain 2016](#)). In this case study, one HEAL fellow working with JSS, Qiratulanne Khan, decided to investigate the needs and experiences of rural women as they relate to sanitation, after she was surprised by how many cases of constipation she saw among her patients. After meeting at JSS, Lillian Walkover worked with her to adapt her insights into a case study for use with structural competency training designed by the Structural Competency Working Group (see [Figure 4.1](#)) and used to teach undergraduate students, clinicians, and global health professionals. Bassem Ghali and Anup Agarwal, both HEAL fellows, built on this work to expand the original research and case study. Together, the authors' work led to the research and trainings described here and will hopefully help others working on similar issues in India and around the world.

Level VI: Policy

Effective policy-level solutions to poor sanitation and open defecation as causes of constipation are few and far between, and this case study does not include examples of action at a policy level. However, while working on a policy level can be daunting from a clinical perspective, clinicians and other health workers have an important role to play in advocacy (for examples of this in action see [Adams 1998](#); [Harris 2017](#)). Policies shape the environment in which patients and providers live and practice, and the resources to which they have access. Indian Prime Minister Narendra Modi has pushed a sanitation agenda in India and is committed to ending the practice of open defecation as part of the Swachh Bharat Abhiyan scheme. As we have seen, simply building toilets and sanitation infrastructure will not stop open defecation in India, as exemplified by the experiences of Nikita and her neighbors in rural central Chhattisgarh. This should not deter advocacy for changes at the policy level. One important piece of this work is the larger policy and other shifts in gender equity that are needed to allow women to feel safe using latrines. JSS is a member of Jan Swasthya Abhiyan, the Indian chapter of the global People's Health Movement. Through this network they contribute to and support a wide variety of health policy demands, including organizing around the right to health ([JSA 2015](#); [PHM 2018](#)).

Conclusion

Nikita's story is a common one in rural Central India. The barriers she faces to healthy defecation—and many of the resources she draws on in her family and community to try to achieve that—are barriers and resources encountered and drawn on by people around the world. Taking a structural analysis, we can see the connections between lack of safe and healthy locations and time in which to stool, larger social factors such as migration and poverty, and structural factors such as gender-based violence and the marginalization of tribal communities. The resources she draws on, including talking with trusted family members, coordinating with family and friends to travel safely, using infrastructure when it is built in a way that works for her life, and taking the advice of first local healers and later JSS staff who provide treatments and insight, may help her to find solutions to her immediate health problems. However, the greater structural arrangements in this region will require much larger efforts to sort out the role of political and social change, the rights of Adivasi people to self-determination, and the complex construction of an evidence base around and options for healthy defecation. Together with her community, and with people at JSS and beyond who are working across levels to change the structural conditions, the social conditions, and the immediate interpersonal and environmental causes of health, Nikita and her story can be part of recognizing and addressing the challenges that women face—and overcome—every day.

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